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Neuroregeneration in the Nucleus Ambiguus After Recurrent Laryngeal Nerve Avulsion in Rats

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Abstract

Objectives: The objective was to investigate neuroregeneration, the origins of newborn cells and the proliferation of neuronal and glial cells in the nucleus ambiguus (NA) after ipsilateral recurrent laryngeal nerve (RLN) avulsion.

Methods: All of the animals received a CM-DiI injection in the left lateral ventricle. Forty-five adult rats were subjected to a left RLN avulsion injury, while 9 rats were used as controls. 5-Bromo-2-deoxyuridine (BrdU) was injected intraperitoneally. Neuron quantification and immunohistochemical analysis were performed in the brain stems at different time points after RLN injury.

Results: After RLN avulsion, CM-DiI labeled neural progenitor cells (NPCs) migrated to the ipsilateral NA and differentiated into astrocytes but not into neurons. In the NA, the neuronal cells re-expressed nestin. Only a small number of neuronal and glial cells in the NA showed BrdU immunoreactivity.

Conclusions: After RLN avulsion, the NPCs in the ependymal layer of the fourth ventricle or central canal are activated, migrate to the lesion in the NA and differentiate exclusively into astrocytes. The newborn neural stem cells in the NA may arise from the mature region neurons. The presence of both cell types in the NA may play a role in repairing RLN injuries.

Keywords

recurrent laryngeal nerve, nucleus ambiguus, neuroregeneration, avulsion

Introduction

Recurrent laryngeal nerve (RLN) injuries frequently result in symptoms of dysphonia, aspiration and dysphagia.[1-4] The incidences of permanent and temporary RLN paralysis after thyroidectomy were 0.9% and 5.1%, respectively.[5] Previous studies have reported that the injury of a peripheral nerve results in a retrograde axon reaction, including neuronal survival and chromatolysis, glial reactions, and the Wallerian degeneration of the distal segment of the axon.[6] Recently, some researchers have reported that central reorganization and neuronal phenotype switches can be induced following a peripheral nerve injury.[7,8] Neurons shift their states from transmitting to regenerative and increase the synthesis of growth-associated proteins and cytoskeleton proteins.[8] Injuries to the peripheral nerve can also cause neural progenitor cells (NPCs) in the ependymal layer to become activated and proliferate.[9] However, the loss of motor neurons in the nucleus ambiguus (NA) caused by vagal/RLN injury is one of the most important factors contributing to the difficulties of treating nerve paralysis.[10] Therefore, it is of great clinical significance to investigate regeneration in the NA after RLN injury.

Recently, neurogenesis in the adult central nervous system (CNS), including the subventricular zone of the lateral ventricle, hippocampal dentate gyrus, and olfactory bulb, has been reported to occur under physiological situations.[11,12] The fourth ventricle and central canal in the brainstem and spinal cord[13,14] are also described as neurogenic regions. Multipotent neural stem/progenitor cells in these neurogenic regions induce self-renewal and can differentiate into neurons, astrocytes, and oligodendrocytes.[15] Injuries[16] and pathological stimuli, such as seizures,[17] stroke,[18] and multiple sclerosis,[19] can activate neurogenesis and even induce NPCs to migrate a long distance to the area of the lesion. NPCs could therefore be used as sources to regenerate motor neurons after vagal/RLN injury. However, when the microenvironment changed due to these injuries and pathological stimuli, mature neurons, and/or glia can also be stimulated to become immature.[20,21]

While compensatory neurogenesis in an injured brain is now well known, it remains unclear whether NPCs can

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migrate to the NA to repair the area of a lesion caused by RLN injury. The aims of the present study were as follows: (1) to investigate whether NPCs in the ependymal layer of the fourth ventricle or central canal are involved in the restoration of the NA after RLN injury, (2) to identify the origins of the newborn cells in the NA following RLN avulsion, and (3) to test the proliferative status of neuronal and glial cells in the NA post-RLN injury.

Materials and Methods

Animals and Surgical Procedures

All animal experiments were conducted in accordance with protocols approved by the local ethical committee on animal research and animal care.

Fifty-four male Sprague-Dawley rats (200-230 g) were deeply, intraperitoneally anesthetized with ketamine (50 mg/kg) and xylazine (10 mg/kg). Then, 20 µl of 0.2% CM-Dil (Molecular Probes, Eugene, Oregon, USA) in DMSO was injected into the lateral ventricle on the left side (0.8 mm posterior, 1.4 mm lateral of the bregma, and 3.7 mm below the dura mater)[22] by stereotaxic injection to label the cells of ependymal layer.

Animals in the injury group were subjected to an avulsion injury (n = 45) of the left RLN 4 days after CM-Dil administration. A vertical skin incision was made in the neck, and the RLN on the left side was carefully exposed under an operating microscope. At the level of the seventh tracheal ring, the proximal RLN was avulsed and removed from the distal RLN by gentle traction using microhemostat forceps. In the control animals (n = 9), skin incisions and soft tissue dissections were performed without injury to the nerves.

Starting the day of the injury, all of the animals received intraperitoneal injections of 50 mg/kg 5-Bromo-2-deoxyuridine (BrdU, Sigma, St Louis, Missouri, USA) dissolved in normal saline twice a day for a maximum of 10 days.

Histological Analysis

At 6 hours, 12 hours, 1 day, 3 days, 5 days, 7 days, 14 days, 21 days, and 28 days after the left RLN injury (n = 5 at every time point), injured and control animals were reanesthetized and perfused transcardially with 37°C saline, followed by 4°C 4% paraformaldehyde in PBS. The brainstems were immediately harvested, postfixed for 2 hours in 4°C 4% paraformaldehyde in PBS, followed by a rinse in PBS, and kept at 4°C in 30% sucrose in PBS for cryoprotection.

The brainstems were embedded in OCT compounds (Sakura, Tokyo, Japan) and cryosectioned at −20°C in sequences of 14 µm + 14 µm + 20 µm, ranging 3.5 mm rostrocaudally.[6] The sections were serially mounted on separate slides. The 14 µm sections were used for immunohistochemistry, and the 20 µm sections were used for neuronal quantification.

To obtain cell counts of the motor neurons, the 20 µm sections were stained with cresyl violet and dehydrated in a graded series of ethanol and xylene prior to mounting in resins. The NA motor neurons that had a clear nucleus and a distinct nucleolus were counted using a wide-field microscope (Leica DM400B, Leica Microsystem GMBH, Mannheim, Germany). The method of Abercrombie[23] was used to avoid errors related to the volume changes of individual neuronal profiles.

For immunohistochemistry, the 14 µm sections were washed 3 times for 15 minutes with PBS and then incubated in PBS containing 10% normal goat serum and 0.3% Triton X-100 (Sigma) for 60 minutes. Sections were incubated at 4°C overnight in PBS containing 0.3% Triton X-100 and primary antibodies, as follows: glial fibrillary acidic protein (GFAP, rabbit, polyclonal, 1:1000; Novus, Littleton, Colorado, USA), nestin (mouse, monoclonal, 1:250; Millipore, Darmstadt, Germany), nestin (rabbit, polyclonal, 1:250; Abcam, Cambridge, UK), doublecortin (DCX, rabbit, polyclonal, 1:250, Cell Signaling, Beverly, Massachusetts, USA), class III β-tubulin (TuJ1, rabbit, monoclonal, 1:500; Covance, Berkeley, California, USA), neuronal nuclei (NeuN, rabbit, polyclonal, 1:100; Novus), Olig2 (rabbit, polyclonal, 1:200; Abcam), ionized calcium binding adaptor molecule 1 (Iba1, rabbit, polyclonal, 1:500; Wako, Osaka, Japan), and ED1 (mouse, monoclonal, 1:1000; Abd Serotec, Oxford, UK).

For the staining of the dividing cell marker BrdU, sections were pretreated by incubation with 2 M HCL for 15 minutes at 37°C; the acid was neutralized by immersing the sections in 0.1 M $Na_2B_4O_7$ for 10 minutes at room temperature. The sections were washed 3 times in PBS, incubated in 0.05% trypsin for 10 minutes at 37°C, followed by a rinse with PBS, and then incubation with antisera for BrdU (mouse, monoclonal, 1:500) at 4°C overnight.

Subsequently, all of the sections were washed 3 times in PBS and then incubated with species-specific secondary antibodies diluted in PBS containing 0.3% Triton X-100 at room temperature for 2 h. The secondary antibodies used included the following: Alexa 488 goat antimouse (1:500; Molecular Probes), Alexa 647 goat antimouse (1:500; Molecular Probes), and Alexa 647 goat antirabbit (1:500; Molecular Probes). The slides were washed 3 times and counterstained with the nuclear marker DAPI (1:500; Molecular Probes). Afterward, the sections were washed and mounted for microscopy. The Leica TCS SP V laser scanning confocal microscope (Leica Microsystem GMBH) was used for all analyses.

Statistical Analysis

For statistical analysis, a 1-way analysis of variance was used, followed by Tukey's posttest (Graph-Pad Software,

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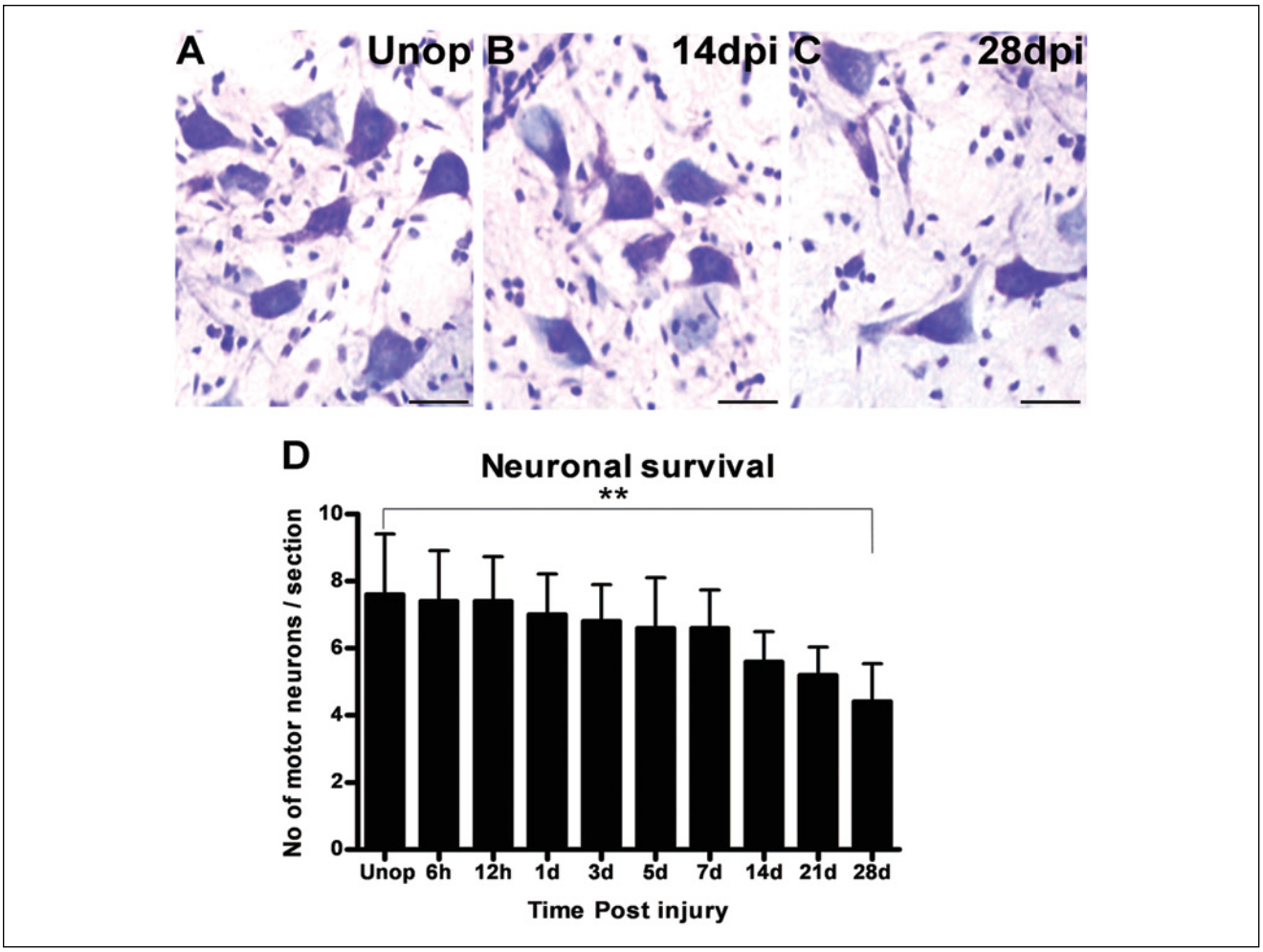


Figure 1. Cresyl violet stained motor neurons in the nucleus ambiguus (NA) projecting into the recurrent laryngeal nerve (RLN) were shown at (B) 14 days and (C) 28 days postinjury, compared with (A) unoperated animals. (D) In contrast to the control animals, a decrease in the number of motor neurons was observed in animals with RLN avulsion injuries at 1 month. Values are mean ± SD. Bars = 10 um. Dpi, days post injury; Unop, unoperated animals. $^{**}P < .01$.

San Diego, California, USA) for the comparison of the cell numbers at each time point. The data are expressed as mean ± SD, and values of $P < .05$ were considered statistically significant.

Results

Quantification of Motor Neurons in the NA Ipsilateral to Injury

In contrast to the control animals, a decrease in the number of motor neurons stained with cresyl violet was observed in animals with RLN avulsion injuries at 1 month (Figure 1, A, B, and C). A massive motor neuron loss (57.93 ± 4.82%) was demonstrated in the NA ipsilateral to RLN injury. The number of motor neurons surviving in the avulsion group was significantly lower than in the control group at 28 days postinjury (Figure 1D).

Glial Reactions in the NA Ipsilateral to Injury

A diverse span of glial reactions with astrocytes, microglia and macrophages was observed in the NA ipsilateral to the injury. GFAP immunoreactivity (a marker for astrocytes) increased from 6 hours postinjury. The maximal immunoreactivity of GFAP was observed at 14 days postinjury, followed by a slight decline at 28 days postinjury (Figure 2, A, B, C, and J), compared with control animals. Iba1 was used as a marker for microglia, and the results indicated a gradual increase in immunoreactivity from 6 hours to 7 days postinjury. At 28 days postinjury, a decrease of Iba1 was observed (Figure 2, D, E, F, and K). ED1 immunoreactivity (a marker for macrophages) was observed at 7 days postinjury and gradually increased past that point until 28 days postinjury, compared with unoperated animals (Figure 2, G, H, I, and L).

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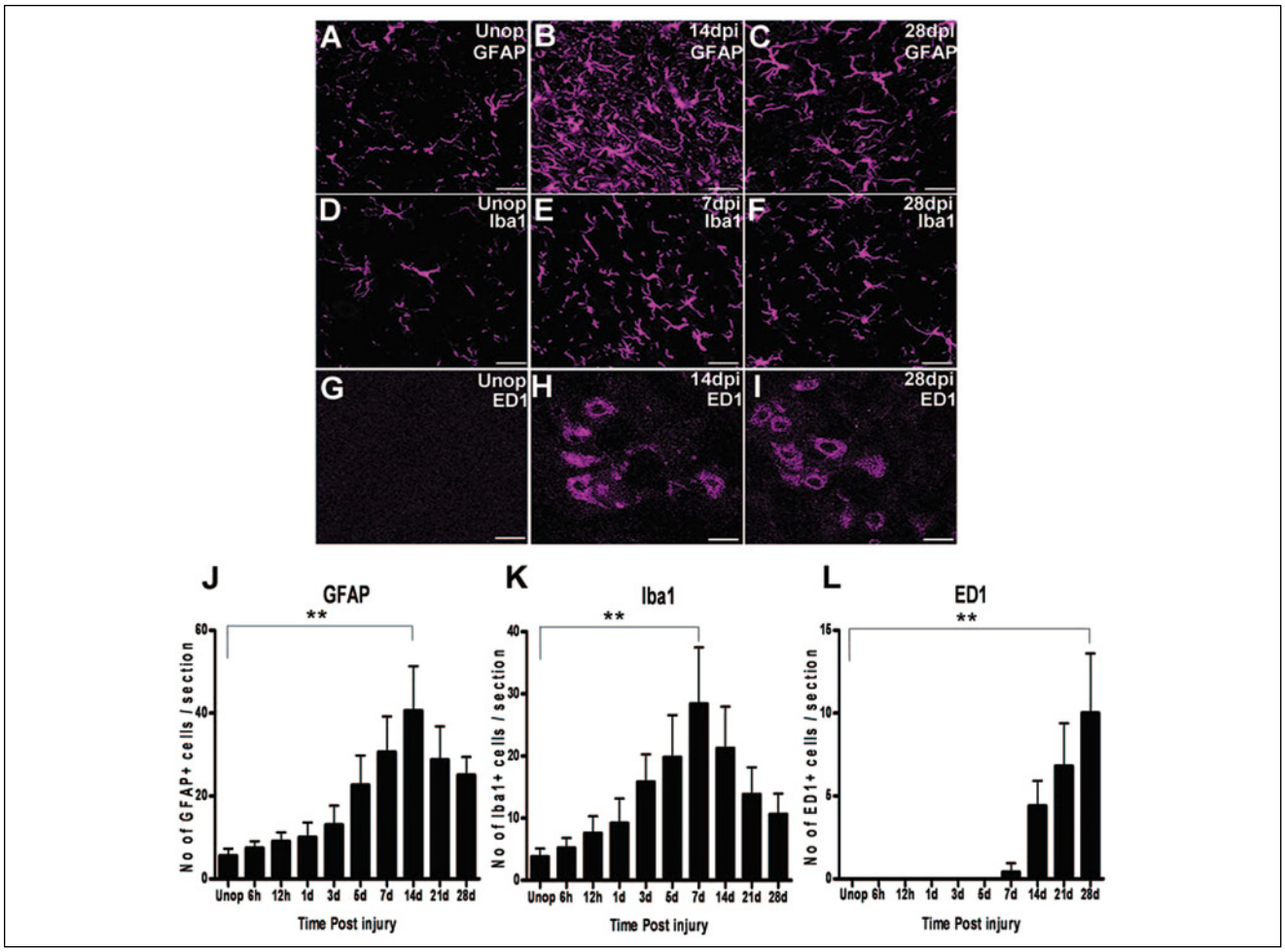


Figure 2. Glial fibrillary acidic protein (GFAP)–positive astrocytes were observed in the nucleus ambiguus (NA) at (B) 14 days and (C) 28 days postinjury, compared with (A) unoperated animals. Iba1-positive microglial were seen in the NA at (E) 7 days and (F) 28 days postinjury, compared with (D) unoperated animals. No ED1-positive macrophages were observed in (G) unoperated animals. Few macrophages were observed at (H) 14 days after nerve avulsion and several macrophages were found at (I) 28 days postinjury. (J) Astrocytes increased from 6 hours postinjury. The maximal immunoreactivity of GFAP was observed at 14 days postinjury, followed by a slight decline at 28 days postinjury, compared with control animals. (K) A gradual increase in immunoreactivity of microglial was observed from 6 hours to 7 days postinjury. At 28 days postinjury, a decrease of Iba1 was observed. (L) The ED1-positive macrophages were observed at 7 days postinjury and gradually increased past that point until 28 days postinjury, compared with unoperated animals. Values are mean ± SD. Bars = 25 um. Dpi, days post injury; unop, unoperated animals. $^{**}P < .01$.

Migration and Differentiation of NPCs From the Ependymal Layer to the NA

To evaluate migration, the cells of ependymal layer were labeled with CM-Dil. BrdU was used to analyze the proliferating cells. BrdU immunoreactivity was obviously expressed in the ependymal cells labeled with CM-Dil after RLN avulsion injury, compared with control animals (Figure 3, A, B, G, H, and I). The CM-Dil-labeled cells were observed in the region between the ependymal layer and the ipsilateral NA from 6 hours to 28 days postinjury, compared with the contralateral side. Some cells in the migratory stream were positive for BrdU, indicating that they were proliferating (Figure 3, C, D, J, K, and L). CM-Dil immunoreactivity was observed in the ipsilateral NA 14 days postinjury, compared with the contralateral side. To characterize the CM-Dil-positive cell, we used markers specific for neuronal cells, astrocytes and oligodendrocytes. The migrating CM-Dil-labeled cell in the NA was colocalized with GFAP but immunonegative for nestin, DCX, TuJ1, NeuN, and Olig2. BrdU immunoreactivity was also expressed in astrocytes in the NA ipsilateral to the injury, and the CM-Dil-labeled cells were colabeled with BrdU and GFAP (Figure 3, E, F, M, N, O, and P).

Nestin Re-expression in the NA Ipsilateral to Injury

Nestin immunoreactivity (a marker for immature cells) was observed prominently in the NA ipsilateral to the injury,

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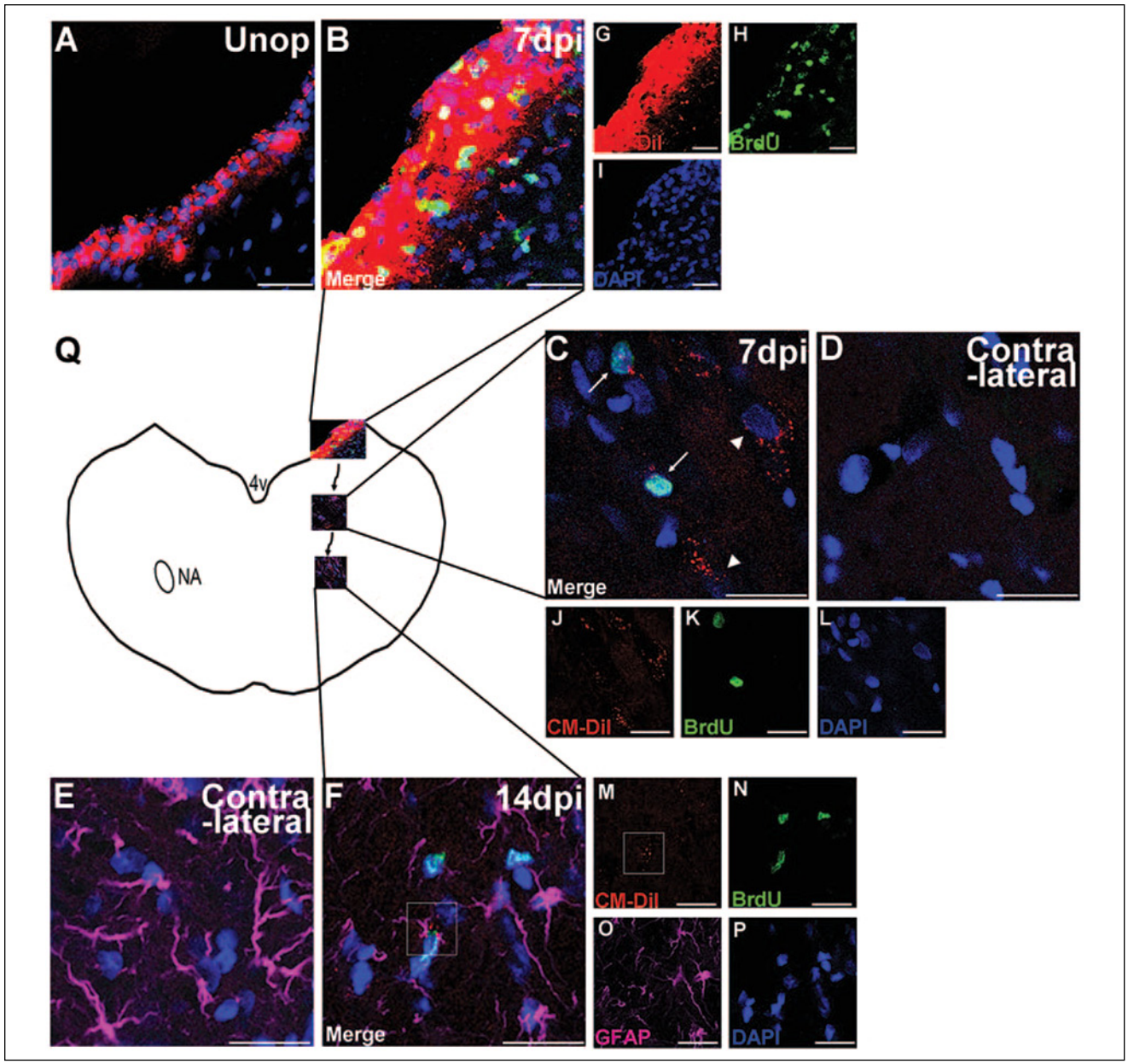


Figure 3. (Q) Schematic illustration showed the routes of the cells of ependymal layers migrating into the ipsilateral nucleus ambiguus (NA) at 7 days after the recurrent laryngeal nerve (RLN) avulsion injury. In contrast to (A) the control animals, (B) BrdU immunoreactivity (green) was obviously expressed in the ependymal cells labeled with CM-DiI (red) after the RLN avulsion injury, indicating that the ependymal cells were proliferating. Merged images of (G) CM-DiI (red), (H) BrdU (green) and (I) DAPI (blue) are shown as (B). (C) The CM-DiI-labeled cells (red) were observed in the region between the ependymal layer and the ipsilateral NA, compared with (D) the contralateral side. Some cells (arrow) in the migratory stream were positive for BrdU (green), indicating that they were proliferating. However, others (arrowhead) showed no immunoreactivity for BrdU. (C) is the merged image of (J) CM-DiI (red), (K) BrdU (green), and (L) DAPI (blue). (F) The CM-DiI-labeled cell (red, white frame) was observed in the ipsilateral NA at 14 days postinjury, compared with the (E) contralateral side. The (M) CM-DiI-labeled (red) cell was colabeled with (N) BrdU (green), (O) GFAP (violet) and (P) DAPI (blue), indicating that the neural progenitor cells migrated into the ipsilateral NA and differentiated into proliferating astrocyte. Bars in A-B, G-I = 40 um, C-F and J-P = 25 um. Dpi, days post injury; NA, nucleus ambiguus; unop, unoperated animals; 4V, fourth ventricle. This figure is available in color online at http://aor.sagepub.com.

compared with controls (Figure 4, A and C). The nestin-positive cells were morphologically similar to neuronal cells. These cells were immunoreactive for TuJ1 (a marker for mature and immature neurons; Figure 4, D, H, I, and J) but were not positive for GFAP (Figure 4, B, E, F, and G). The quantitative data on nestin-positive cells in the NA

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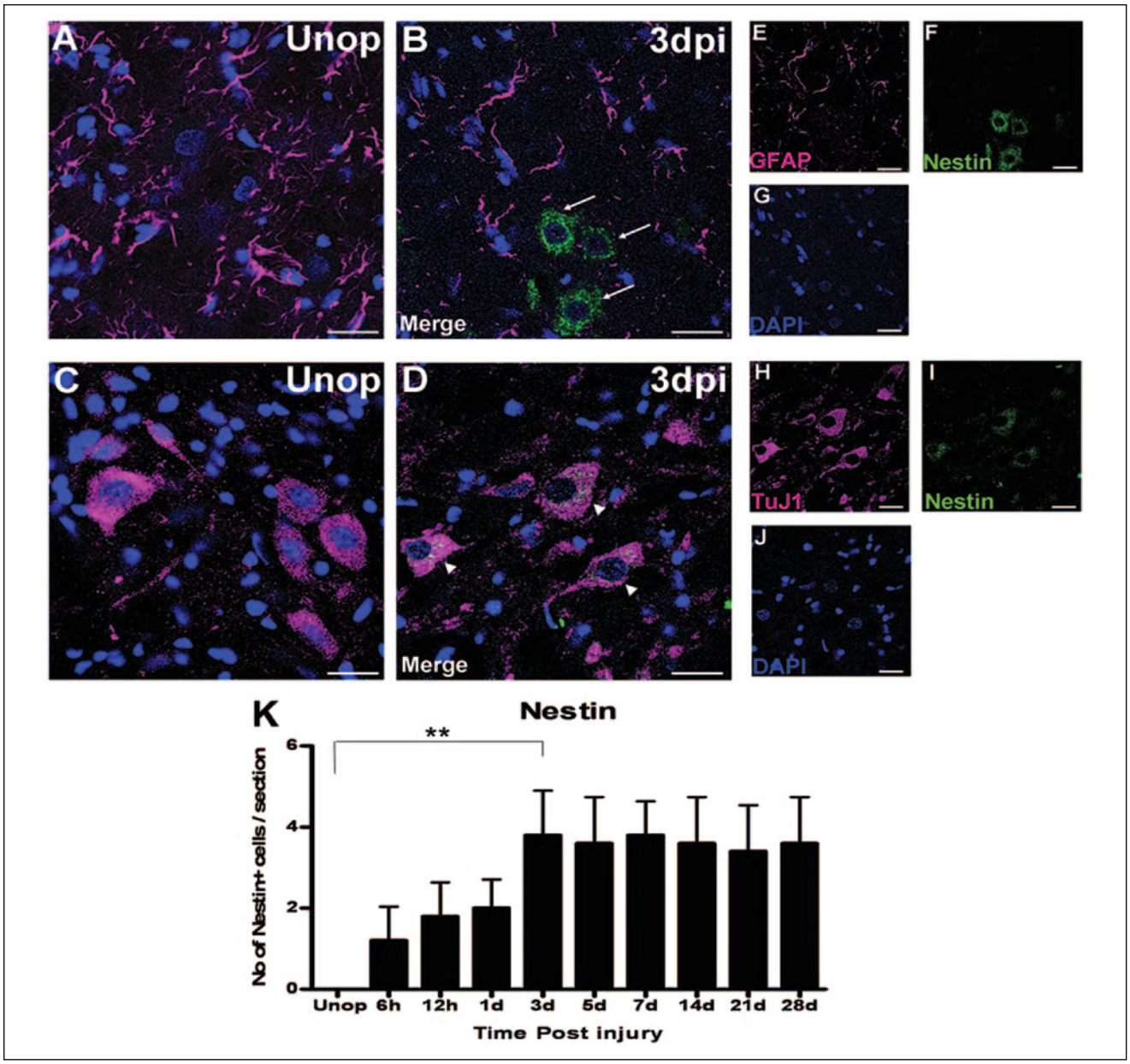


Figure 4. Expression of Nestin in the ipsilateral nucleus ambiguus (NA) and in controls (A, C) at 3 days post the recurrent laryngeal nerve (RLN) avulsion injury. Nestin-positive cells (green, arrow) showed (B) no immunoreactivity for GFAP (violet), but were (D) immunoreactive for TuJ1 (violet). (B) is the merged image of (E) GFAP (violet), (F) Nestin (green), and (G) DAPI (blue). (H) TuJ1 (violet), (I) nestin (green), and (J) DAPI (blue) are merged as (D). These findings indicated that the nestin-immunoreactive cells were neuronal cells but not astrocytes. (K) The quantitative data on nestin-positive cells in the NA ipsilateral to the injury showed an increase from 6 hours to 3 days postinjury and remained almost the same at later survival times. Values are mean ± SD. Bars = 25 um. Dpi, days post injury; unop, unoperated animals. $^{**}P < .01$. This figure is available in color online at http://aor.sagepub.com.

ipsilateral to the injury (Figure 4K) showed an increase from 6 hours to 3 days postinjury and remained almost the same at later survival times. To evaluate the neurogenesis and migration, DCX was used in the NA; however, no immunoreactivity was demonstrated.

Proliferation in the NA Ipsilateral to the Injury

BrdU was analyzed as a proliferative marker for NA. No BrdU immunoreactivity was observed in controls (Figure 5, A and C). Some of the GFAP-positive cells expressed BrdU

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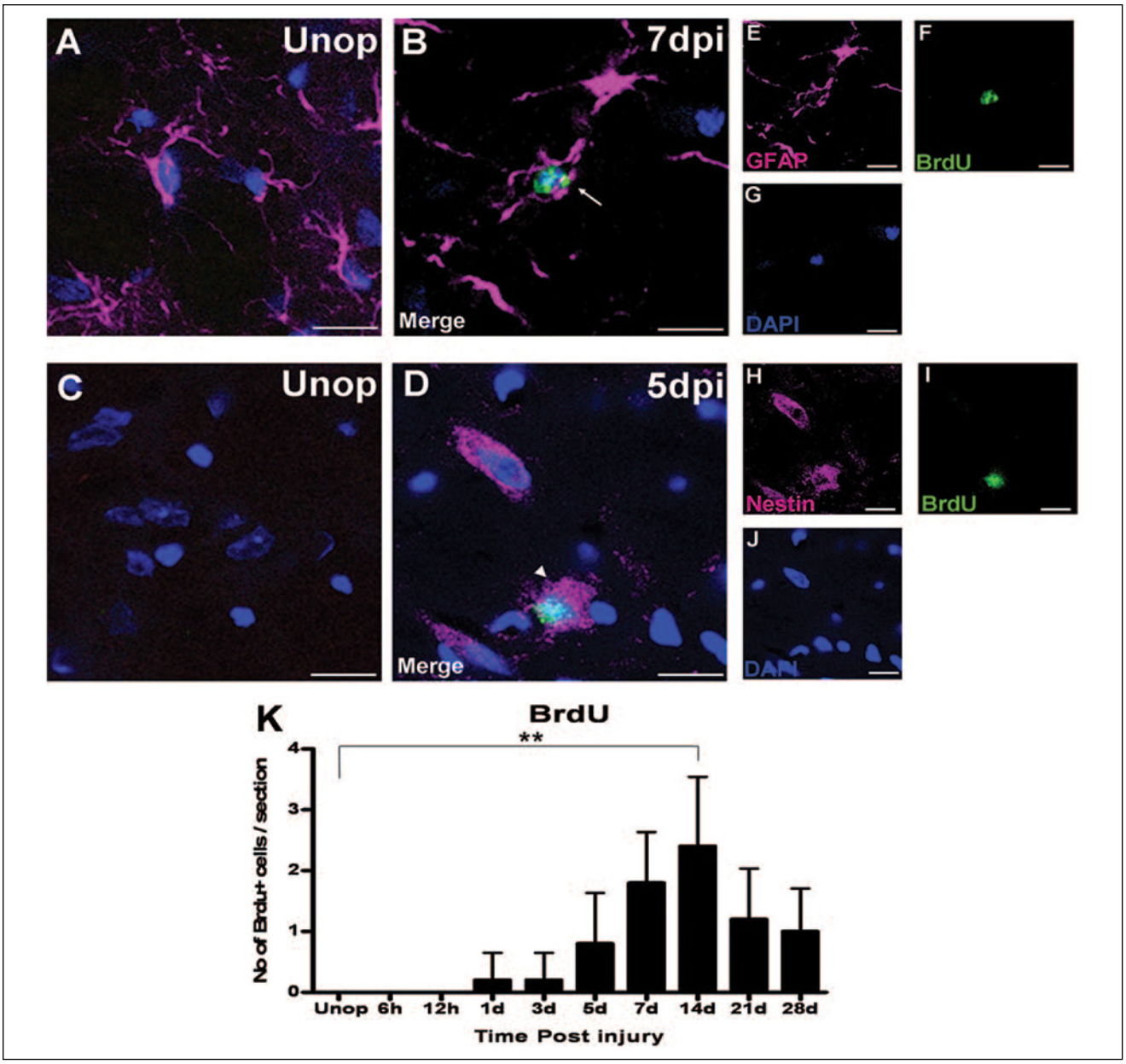


Figure 5. (A, C) No proliferating cell was observed in the nucleus ambiguus (NA) in control animals. (B) Both GFAP (violet) and BrdU (green) immunoreactive cell (arrow) were observed in the ipsilateral NA at 7 days postinjury. (B) is the merged image of (E) GFAP (violet), (F) BrdU (green), and (G) DAPI (blue). (D) Nestin-immunoreactive (violet) cell was positive for BrdU (green, arrowhead) in the ipsilateral NA at 5 days postinjury. (H) Nestin (violet), (I) BrdU (green), and (J) DAPI (blue) are merged as (D). These findings indicated that both astrocyte and neuronal cell in the NA ipsilateral to injury could be proliferating. (K) Quantification of the BrdU-positive cells showed that the proliferating cells gradually increased from 1 day to 14 days after the injury and declined from 21 to 28 days after the injury. Values are mean ± SD. Bars = 10 um. Dpi, days post injury; unop, unoperated animals. $^{**}P < .01$. This figure is available in color online at http://aor.sagepub.com.

(Figure 5, B, E, F, and G), although only 1 cell was both Nestin- and BrdU-immunoreactive (Figure 5, D, H, I, and J). Quantification of the BrdU-positive cells showed that the proliferating cells gradually increased from 1 day to 14 days after the injury and declined from 21 to 28 days after the injury (Figure 5K).

Discussion

Neural injury or disease can trigger compensatory neurogenesis. Moreover, endogenous NPCs can be used to repair the region of the lesions.[24] Pathological processes can stimulate the proliferation of NPCs and redirect them toward

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migration and differentiation, away from the rostral migratory stream to the olfactory bulb or the site of damage.[25] However, mature neurons can also be stimulated by injury or disease to reverse their states and to re-express the immature neural cell marker Nestin.[20,26] It has been shown that the mature motor neurons in the NA may demonstrate this juvenilization potential postvagal/RLN injury.[10] However, whether the NPCs in the ependymal layers are activated and migrate to repair the ipsilateral NA remains unclear.

When a peripheral nerve injury occurs, the neuronal cell bodies are induced to degenerate. The variable degrees of neuronal degeneration depend on the injury type,[24] region of neural injury,[27] extent of damage,[9] animal species, age,[28] and so on. As previous studies have reported,[2-4,29] decreased motor neuron survival numbers were observed in the NA post-RLN avulsion injury in our study. Reactive microglial cells were observed around the RLN neurons, followed by astrocytes migrating to the region of the lesion. These results indicate a typical inflammatory response pattern after peripheral motor nerve injuries.[6] Furthermore, ED1-positive macrophages were also observed in the NA, which may be the site of microglial transformation.[9,28] The appearance of macrophages may help to explain neuronal cell death. These findings suggest that the glial reactions caused by RLN avulsion may be helpful for neuroregeneration in the NA.

The notion that ependymal layer cells are neural stem cells has been proven in spinal cord injury, inflammatory disease,[14,19] peripheral nerve injury,[2] and so on. Neurogenesis caused by neural damage occurs following a specific sequence of events: NPC activation, proliferation and migration, followed by the survival and maturation of the newborn neurons.[30] According to this pattern, we used BrdU, a marker that can be incorporated into newly synthesized DNA,[31] as the identifying marker to characterize proliferating NPCs. The results showed that the BrdU+/CM-Dil+NPCs increased in the ependymal layer of the fourth ventricular or central canal in the avulsion model, compared with control animals, from 6 hours to 28 days postinjury. This result indicates that the activation and proliferation of the NPCs in the ependymal layer occur due to the RLN avulsion injury. Proliferation in neurogenic regions in response to brain/nerve injuries has been demonstrated by previous studies.[9,32] The CM-Dil-labeled migratory stream cells were also observed between the ependymal zone and the ipsilateral NA in this study. In the migratory stream, some BrdU-positive cells were found to be proliferating. The result suggests that NPCs migrate to the region of the lesion, induced by the avulsion injury. Takaoka et al[10] reported that no flow of migratory cells was observed in the vagal nerve resection model. However, their observation period was shorter than ours, and the degree of nerve damage was more severe in our observations. Furthermore, as the CM-Dil+/BrdU+/GFAP+ cells were observed in the ipsilateral NA at 14 days postinjury, this finding indicates that the proliferating NPCs had differentiated into astrocytes but not into neurons or oligodendrocytes. Previous studies have reported the astroglial differentiation of NPCs following different CNS lesions.[9,14] However, neuronal differentiation has also been reported.[18,33] Furthermore, this induced neurogenesis could be temporal. NPCs expressed neuronal markers at the beginning of the experiment but expressed glial markers later.[34] In our study, astroglial differentiation may be due to the niche changing in accordance with RLN injury. The response of the NPCs to neural damage is mediated by endogenous factors, such as cytokines and chemokines, which are released by the inflammatory response at the injured region, by the productions of neurotrophic factors, growth factors and morphogens and by the extracellular matrix.[24,35] Some factors[36] can induce glial lineage differentiation by inhibiting neuronal differentiation in NPCs. Therefore, the successful repair in vivo postinjury depends on the appropriate modulation of the multiple facets of NPC biology.[25] Indeed, limited CM-Dil-labeled cells were observed in the NA postinjury, indicating that both glial and neuronal differentiation may be suppressed in vivo postinjury.[35] The repair capability in the NA is limited after RLN injury. Neurotrophic factors may be necessary to promote the regeneration in the NA after RLN injury.

Nestin, an intermediate filament protein,[37] has been reported to be expressed in neuronal and glial precursor cells[26] and reactive astrocytes.[38] The upregulation of nestin may be a result of new cellular division and can also demonstrate the re-expression of nestin within existing glia.[39] Double-labeling studies indicated that nestin-positive cells were not immunoreactive for GFAP, the astrocytic marker; however, nestin was found to be colocalized with the neuronal marker TuJ1. Furthermore, the nestin-positive cells were not immunoreactive for CM-Dil but had a neuronal morphology. These results suggest that regional neurons in the NA produce nestin in response to ipsilateral RLN avulsion injury and that they showed an immature phenotype, unlike migratory newborn cells. Mature neurons have been previously reported to be positive for nestin in several studies, such as ischemic brain injury[20] and remote cortical injury.[26] A previous study[10] also demonstrated that mature neurons re-expressed nestin in the NA after vagal nerve resection, and our findings support this view. We speculate that this reversion may be induced by the changed microenvironment in the NA after RLN injury. Nestin is not expressed in mature neurons. However, endogenous factors such as cytokines and chemokines appear around the NA after RLN injury. Mature neurons are mediated to become immature and express nestin again. Moreover, the astrocyte-differentiated NPCs in the NA may also produce the necessary inflammatory factors to promote this re-juvenilization.

Proliferation in the NA on the side of the lesions was investigated using the marker BrdU. Previous studies have demonstrated BrdU-immunoreactivity at the site of lesional

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damage after various injuries;[9,40] however, others have not.[10,26] Our study showed that most BrdU-positive cells are astrocytes, and only 1 cell was observed to be colocalized with the neural stem cell marker nestin at 5 days postinjury. This result indicates that the mitotic activity of nestin-positive cells that appeared in the NA postinjury was not high enough to be observed or that these cells may show very slow mitotic activity.[10] Moreover, the proliferation of astrocytes may be due to the inflammatory response in the NA caused by RLN injury.

Conclusion

We considered that neuroregeneration occurs in the NA post-RLN avulsion injury. In this study, we first confirmed that the migratory NPCs appeared in the NA and that they differentiated only into astrocytes. Moreover, the neural stem cells in the NA may come from the region mature motor neurons. The detailed migratory process of NPCs stimulating endogenous neurogenesis post-RLN injury should be further studied. These findings will contribute to future possibilities for the treatment and neurofunctional recovery of laryngeal paralysis.

Declaration of Conflicting Interests

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